

Htet Myat Aung (Alexander E. Hong)

• GitHub: <https://github.com/AlexHongHMA>



PROFESSIONAL SUMMARY

Highly motivated Automation Engineering graduate with extensive experience in artificial intelligence, machine learning, and robotics research. Demonstrated expertise in developing advanced computer vision systems, autonomous vehicle control, and deep learning applications. Award-winning researcher with publications in prestigious IEEE conferences and proven track record of academic excellence. Seeking opportunities to apply innovative AI/ML technologies in real-world applications.

RESEARCH EXPERIENCES

Student Researcher | OPTIMISED 3D-CNN FOR REALTIME INFRARED NATURAL GAS LEAK CLASSIFICATION

Bangkok, TH
2024 June – Present

- Developed a high-accuracy (99.75%) 3D CNN model for methane gas leak detection.
- Applied Running Average, Gaussian Mixture Models (GMM), Moving Average, and Background Subtraction techniques.
- Implemented smaller image sizes (Half Size and Quarter Size) and received above 99 percent accuracy for efficient deployment.
- **Co-Winner of Best Paper Award at 13th International Conference on Information and Communication Technology Bandung (Hybrid) - IcoICT 2025 (IEEE Xplore) - July 30-31, 2025 with Asst. Prof. Dr. Sarawan Wongs**
- Currently developing quantization techniques for the award-winning 3D CNN model with Asst. Prof. Dr. Sarawan Wongs for journal publication

Student Researcher | CODIA Lab, KMUTT & Panasonic Singapore SLAM and Computer Vision Systems Development

Bangkok, TH
2024 Feb – 2024 July

- Developed SLAM Sparse Bundle Adjustment algorithms in C.
- Enhanced algorithm efficiency for edge devices.

Student Researcher | COSIN Lab, INC Department Autonomous Vehicle Control Systems

Bangkok, TH
2023 Nov – 2024 Jan

- Developed ROS2-based autonomous navigation and control systems for competitive robotics applications
- Integrated micro-ROS framework with IMU sensor data to achieve enhanced navigation precision and reliability
- Contributed to robotics competition preparation and system optimization

PROFESSIONAL EXPERIENCES

Full Time Machine Learning Researcher | Maintenance Technology Center (MTC) Non-Destructive Detection Technology Lab

Bangkok, TH
2025 Sept – Present

- Currently developing advanced machine learning solutions for non-destructive testing applications
- Collaborating with Prof. Cherdpong Jomdech on cutting-edge detection technology research
- Contributing to innovative maintenance technology solutions for industrial applications

Machine Learning Intern | Maintenance Technology Center (MTC) Non-Destructive Detection Technology Lab

Bangkok, TH
2025 June – 2025 Aug

- Developed both supervised, unsupervised and semi-supervised machine learning algorithms for acoustic detection based non-destructive detection systems
- Won First Runner-Up Award in "Smart Infrastructure for Smart City" competition by Metropolitan Waterworks Authority
- **Listed in KMUTT Hall of Fame (In Thai Language News) for outstanding achievement in Smart Infrastructure development**
- Applied AI/ML techniques to solve real-world infrastructure challenges

Junior Machine Learning Intern | Large Language Model (LLM) Intern HoneyHunt.AI,

Bangkok, TH
2024 Dec – 2025 Feb

- Enhanced automated resume parsing and intelligent job matching systems utilizing advanced LLM APIs
- Improved algorithm scalability and matching accuracy for production-level deployment
- Contributed to product development in AI-powered recruitment technology

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PUBLICATIONS & CONFERENCES

- **13th International Conference on Information and Communication Technology Bandung (Hybrid) - IcoICT 2025 (IEEE Xplore)** - "Optimized 3D-CNN for Real-Time Infrared Natural Gas Leak Classification" – DOI: [10.1109/ICoICT66265.2025.11192973](https://doi.org/10.1109/ICoICT66265.2025.11192973) - **Best Paper Award Winner** **2025 July 30-31**
- **16th Instrumentation, Control and Automation Senior Project Conference - ICA-SP Con 2025** - "Natural Gas Methane Leak Detection Using an Infrared Camera" - **Best Presentation Award Winner** **2025 April**

AWARDS & ACHIEVEMENTS

- **Co-winner of Best Paper Award with Asst. Prof. Dr. Sarawan Wongsu - IcoICT 2025 (IEEE Xplore)** **July 30-31, 2025**
- **Best Presentation Award - ICA-SP Con 2025** **April 2025**
- **First Runner-Up - Smart Infrastructure for Smart City Competition** **July 30th, 2025**
- **KMUTT Hall of Fame - Listed for outstanding achievement in Smart Infrastructure development** **July 31st, 2025**
- **INC Department Outstanding Student Scholarship** **Winter Semester 2023**
- **INC Department Outstanding Student Scholarship** **Winter Semester 2024**

LEADERSHIP & ACTIVITIES

FIRST GLOBAL CHALLENGE 2019

Mentor

Dubai, UAE
2019 June – 2019 Oct

- Provided technical guidance and strategic mentorship to Myanmar national robotics team
- Achievement: Led team to win prestigious Judges Award at international robotics competition
- Fostered STEM education and international collaboration in robotics

LABORATORY OF ROBOTICS AND ARTIFICIAL INTELLIGENCE (L.O.R.A)

Robotics Researcher and Founder

Yangon, MM
2019 Dec – 2021 June

- Founded and led comprehensive robotics research laboratory focusing on practical applications
- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic in Myanmar
- Managed multiple concurrent research initiatives and mentored junior researchers.

EDUCATION

Bachelor of Engineering in Automation Engineering

GRADUATED 2025

King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, Thailand

GPAX: 3.66/4.00

Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera

Department: Control System and Instrumentation (INC)

TECHNICAL SKILLS

Programming Languages: Python, C, Embedded C, C++

Frameworks & Tools: ROS2, TensorFlow, PyTorch, OpenCV

AI/ML Technologies: Large Language Model APIs (OpenAI, DeepSeek, Gemini, Claude)

Core Competencies:

- Machine Learning & Deep Learning
- Computer Vision & Image Processing
- Robot Kinematics & Control Systems
- Simultaneous Localization and Mapping (SLAM)
- Embedded Systems Development
- Signal Processing & Data Analysis